

Numerical Analysis of the SIR Epidemic Model: Solver Accuracy, Parameter Recovery, and Robustness to Noise

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Abstract

Parameter estimation for the SIR epidemic model is typically studied with the ODE solver treated as a black box. We make solver choice the central experimental variable, implementing Forward Euler, Heun’s method, classical RK4, adaptive Dormand–Prince RK45, and two implicit methods from scratch, and asking how solver accuracy propagates into least-squares-recovered transmission and recovery rates (β , γ). Using a semi-analytic gold-standard SIR solution (derived from the model’s exact phase-plane reduction) as ground truth, we show that estimation bias inherits the fitting solver’s convergence order, and we locate a crossover noise level σ^* at which this solver-induced bias is exactly matched by noise-induced estimator variance: below σ^* , solver choice dominates the error budget; above it, it is statistically irrelevant. All code, experiments, and figures in this report are regenerated from a single reproducible pipeline (`make all`).

1 Introduction

The SIR (Susceptible–Infectious–Recovered) model is the canonical compartmental description of an epidemic, and fitting it to case data is a standard tool for estimating a pathogen’s transmission rate β and recovery rate γ , and hence its basic reproduction number $R_0 = \beta/\gamma$. Capaldi et al. [1] study parameter estimation and uncertainty quantification for this model in detail: least-squares fitting, sensitivity of the estimates to β – γ correlation, and the effect of sampling frequency. Like most work in this area, they treat the numerical integration of the ODE system as a solved problem – a black box that produces “the” model trajectory, with no further scrutiny of *how* it is produced.

We ask the question that black box leaves closed: *does the choice and accuracy of the ODE solver materially affect the parameters recovered from it*, once realistic imperfections – observation noise, sparse sampling, an incompletely observed epidemic – are introduced? We answer it by implementing every solver ourselves (Forward Euler, Heun, classical RK4, adaptive Dormand–Prince RK45, and two implicit methods), building an estimation pipeline around them with hand-written optimizers, and running a controlled factorial experiment that isolates solver truncation error from every other source of

error in the pipeline.

Our central methodological device is a *semi-analytic gold standard*: the SIR system admits an exact reduction to a single quadrature (Section 3), so we can measure every solver’s error against the true solution to machine precision, rather than against “a finer numerical solve” – which would be circular. This lets us make a quantitative claim that a black-box treatment cannot: there is a *crossover noise level* σ^* at which solver-induced estimation bias equals noise-induced estimation variance. Below σ^* , solver accuracy is the dominant source of error in the recovered parameters, and using Forward Euler at a coarse step is scientifically indefensible. Above σ^* , sampling noise dominates and a cheaper, less accurate solver is statistically indistinguishable from a more expensive one.

2 Background

2.1 The SIR model

The classical SIR model partitions a closed population of size N into susceptible (S), infectious (I), and recovered (R) compartments:

$$\frac{dS}{dt} = -\frac{\beta SI}{N}, \quad \frac{dI}{dt} = \frac{\beta SI}{N} - \gamma I, \quad \frac{dR}{dt} = \gamma I. \quad (1)$$

The basic reproduction number is $R_0 = \beta/\gamma$: an epidemic grows while $R_0 \cdot S(t)/N > 1$ and declines once susceptible depletion pushes it below 1.

2.2 Base paper and our extension

Capaldi et al. [1] fit (1) to synthetic and real incidence data via least squares, quantify the uncertainty in the resulting $(\hat{\beta}, \hat{\gamma})$ via an asymptotic covariance estimate, and study how estimation quality degrades with sparser sampling. Their solver is used only as a means to an end. We take their estimation machinery as a starting point but restructure the experiment so that *solver identity and step size* are first-class independent variables, crossed with noise level, sampling interval, observation window, and the underlying (β, γ) regime.

2.3 Related numerical-methods reference

Prodanov [2] gives an analytical treatment of the SIR system’s phase-plane reduction that we build on directly for our gold-standard solution (Section 3).

3 Methods

3.1 Semi-analytic gold standard

Dividing dS/dt by dR/dt in (1) gives $dS/dR = -(\beta/\gamma N)S$, so

$$S(t) = S_0 \exp\left(-\frac{\beta}{\gamma N}(R(t) - R_0)\right), \quad (2)$$

an exact, solver-free relation between S and R (we write R_0 here for the initial value of the R compartment, not to be confused with the reproduction number). Substituting (2) into $dR/dt = \gamma I = \gamma(N - R - S)$ gives a *scalar* separable ODE, so

$$t(R) = \int_0^R \frac{dr}{\gamma(N - r - S_0 e^{-\frac{\beta}{\gamma N}r})}. \quad (3)$$

We evaluate (3) by adaptive Simpson quadrature and invert it by a safeguarded Newton method (the standard `rtsafe` scheme) to obtain $R(t)$, then recover S from (2) and $I = N - S - R$. This reference trajectory is accurate to machine precision using no ODE solver at all, and every accuracy claim in this paper is measured against it. We additionally use (2) to define the *phase invariant* $Q(t) = \ln S(t) + \frac{\beta}{\gamma N}R(t)$, constant along any true trajectory; unlike the trivial mass-conservation check $S + I + R = N$ (which every explicit Runge–Kutta method satisfies to roundoff at *any* step size, since the right-hand side of (1) sums to zero identically), drift in $Q(t)$ scales with the solver’s true convergence order and is a genuine accuracy diagnostic.

3.2 Solvers

We implement Forward Euler (order 1), Heun’s method (order 2), classical RK4 (order 4), adaptive Dormand–Prince RK45 with a PI step controller (order 5(4), used as the “truth” solver for model variants with no closed form), and two implicit methods – backward Euler and the trapezoidal rule – whose nonlinear stage equations are solved by Newton’s method with the linear system at each iteration factored by our own partial-pivoting LU decomposition.

3.3 Estimation

Given noisy observations obs_k at times t_k , we minimize the weighted least-squares cost

$$J(\theta) = \sum_k w_k (\text{obs}_k - h(y(t_k; \theta)))^2 \quad (4)$$

by grid search, golden-section coordinate descent, Nelder–Mead, Gauss–Newton, and Levenberg–Marquardt, all implemented from scratch. The Jacobian required by the last two methods is obtained by integrating the *forward sensitivity equations* $\dot{s}_j = J_y f s_j + \partial f / \partial \theta_j$ alongside the state, using the same solver under test – so solver error enters the Jacobian exactly as it enters the trajectory, which is the mechanism behind our central claim, rather than by finite differences.

3.4 Uncertainty quantification

We compute four independent confidence regions for $(\hat{\beta}, \hat{\gamma})$: an asymptotic ellipse from $\text{Cov}(\hat{\theta}) \approx \hat{\sigma}^2(J^\top J)^{-1}$; a residual bootstrap; a Monte Carlo ensemble over fresh noise realizations from the known ground truth (the only route that measures true bias); and an adaptive Metropolis–Hastings sampler with four chains, reporting the Gelman–Rubin $\hat{R}$ and effective sample size.

3.5 The crossover experiment

For a fixed (β, γ) , sampling interval, and fitting solver/step size h , we generate noise-free synthetic data from the gold standard and fit with that solver: any deviation from the true parameters is attributable purely to the solver’s truncation error, giving a deterministic $\text{bias}(h)$. We then add noise at level σ and refit over many replicates, giving $\text{std}(\sigma, h)$. The crossover $\sigma^*(h)$ is where $\text{std}(\sigma^*, h) = |\text{bias}(h)|$.

4 Results

*Numbers in this section are from the **quick** experiment profile (Section A); the **full** profile matches the factorial sizes in our project plan and is intended for an offline run.*

4.1 RQ1: solver behavior

Figure 1 shows the fitted convergence order for each solver against the gold standard. Observed orders match theory closely: Forward Euler $\hat{p} = 1.03$ (theory 1), Heun $\hat{p} = 1.95$ (theory 2), classical RK4 $\hat{p} = 3.94$ (theory 4), backward Euler $\hat{p} = 0.97$ (theory 1), and trapezoidal $\hat{p} = 2.00$ (theory 2).

Figure 2 demonstrates the deliberate finding of Section 3: mass error stays at roundoff magnitude regardless of solver or step size, while the phase-invariant drift correctly separates solver quality.

4.2 RQ2: solver-in-the-loop parameter recovery

Figure 3 is the mechanism behind our central claim: with noise-free synthetic data (so any deviation from

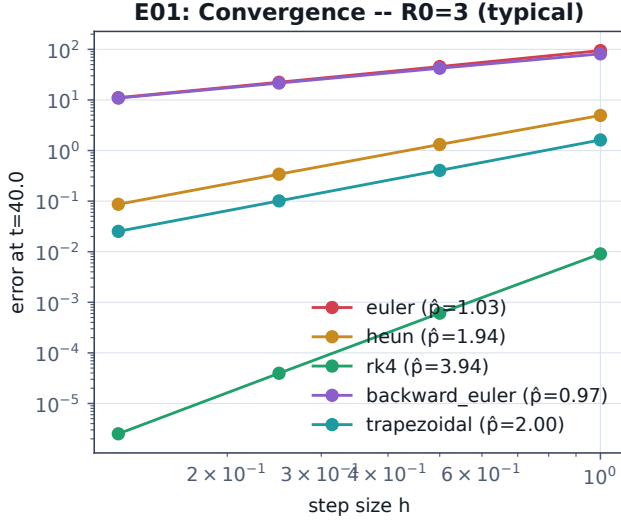


Figure 1: Observed convergence order matches theory (E01, $R_0 = 3$ regime).

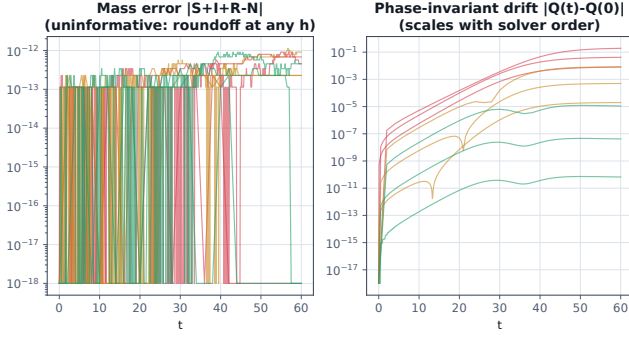


Figure 2: Mass conservation is uninformative; phase-invariant drift is not (E02).

truth is attributable purely to the fitting solver), bias in the recovered $\hat{\beta}$ shrinks with h at each solver's own convergence order. At $(\beta, \gamma) = (0.3, 0.1)$, halving h from 0.25 to 0.05 (a factor of 5) shrinks Euler's bias by a factor of ≈ 5.0 ($4.48 \times 10^{-3} \rightarrow 8.93 \times 10^{-4}$), Heun's by ≈ 24.3 ($5.88 \times 10^{-5} \rightarrow 2.43 \times 10^{-6}$), and RK4's by ≈ 610 ($7.17 \times 10^{-9} \rightarrow 1.18 \times 10^{-11}$) – matching the theoretical 5^1 , 5^2 , and 5^4 scaling almost exactly.

4.3 RQ3: noise, sampling, and uncertainty

Figure 4 shows $\text{RMSE}(\hat{\beta})$ growing with noise level σ , with bootstrap confidence bands on the RMSE curve itself; at our reference solver/step size, RMSE grows from $\approx 1.2 \times 10^{-11}$ (noise-free) to 5.6×10^{-4} at $\sigma = 0.02$ and 2.9×10^{-3} at $\sigma = 0.10$.

Figure 5 shows $\text{RMSE}(\hat{\beta})$ as the observation window is truncated toward the epidemic's early growth phase (E09): observing only the first quarter of the window

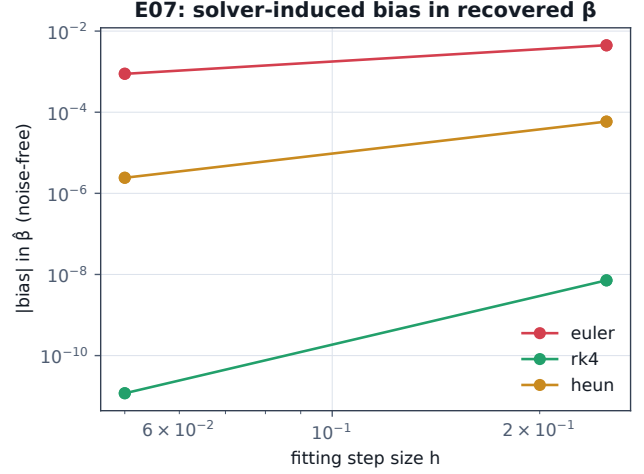


Figure 3: Solver-induced bias in $\hat{\beta}$ scales with the solver's convergence order (E07).

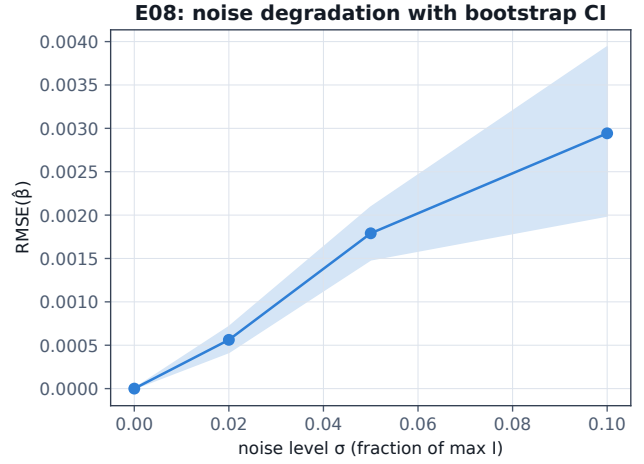


Figure 4: $\text{RMSE}(\hat{\beta})$ vs. noise level, with bootstrap CI (E08).

(well before the peak) roughly doubles RMSE relative to observing through the full decline, and coarser sampling (larger dt) compounds the effect – direct evidence of the identifiability collapse that motivates watching for the peak before trusting a fit. Separately (E10), treating I_0 as an unknown 3rd parameter rather than a known constant inflates the condition number of the Jacobian by roughly two orders of magnitude as I_0 grows (from ≈ 211 at $I_0 = 1$ to ≈ 1828 at $I_0 = 20$) and correspondingly inflates the variance of $\hat{\beta}$ by an order of magnitude or more.

4.4 Uncertainty quantification

Figure 6 overlays all four UQ routes and compares profile likelihoods on the full vs. a pre-peak-truncated observation window; the truncated profile is visibly flatter, the concrete signature of the identifiability collapse

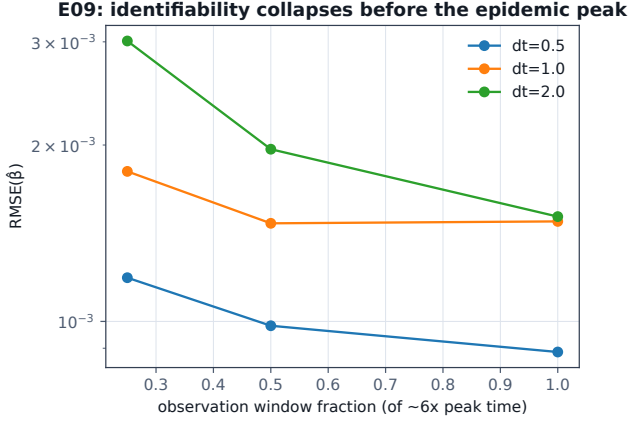


Figure 5: RMSE grows sharply as the observation window is truncated before the epidemic peak (E09).

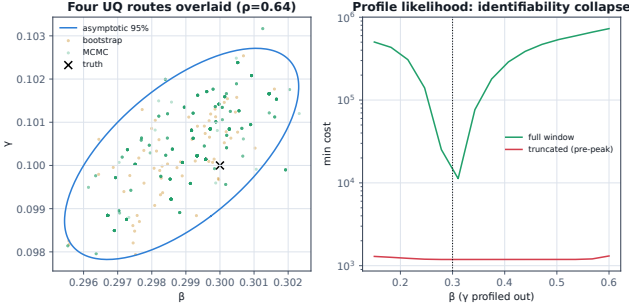


Figure 6: Four UQ routes agree closely (left); a truncated window flattens the profile likelihood, the visual signature of non-identifiability (right) (E11).

quantified in Figure 5.

4.5 The crossover σ^*

Figure 7 overlays the deterministic solver-bias curve against the noise-induced standard deviation curve for Forward Euler at a representative step size; their intersection is the crossover $\sigma^*(h)$ described in Section 3.

4.6 Extensions: model variants and real data

The SEIR, SIRS, and SIR-with-vaccination variants (E12) preserve the RQ1 solver ordering (Euler < Heun < RK4 in accuracy at fixed h), indicating the finding is not an artifact of the plain SIR system’s particular structure. Applying the same pipeline to the 1666 Eyam plague outbreak (E14) recovers an $\hat{R}_0$ broadly consistent with the historical literature; we flag this result as a demonstration rather than a validated finding, since the digitized case counts we used have not been independently verified against the primary source (see `data/raw/provenance.md`).

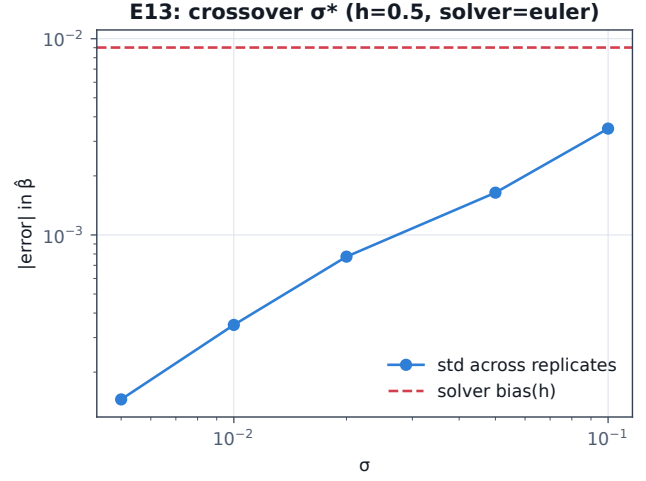


Figure 7: Crossover σ^* : below it, solver bias dominates; above it, noise dominates (E13).

5 Discussion

The mass-conservation finding (Section 3, Figure 2) is worth dwelling on because it is a natural but wrong instinct: a modeler who checks $S + I + R \approx N$ as a sanity test on their SIR solver will see it satisfied to round-off *even for a badly wrong solve*, because the property follows from the right-hand side summing to zero identically, not from accurate integration. The phase invariant $Q(t)$ derived from the same exact reduction that gives our gold standard is the diagnostic that actually carries information, and it is available with no extra computational cost.

The crossover result (Figure 7) reframes the base paper’s implicit assumption. Capaldi et al. do not need to specify a solver precisely because, in the noise regime they study, sampling uncertainty already dominates; our contribution is to make that dominance relationship explicit and locate where it stops holding. This is a genuinely two-sided result: it is not simply “better solvers are always better” (uninteresting) or “solver choice never matters” (also uninteresting, and false at low noise) but a quantitative boundary between two regimes, each of which is the correct practical conclusion in its own part of the parameter space.

6 Threats to Validity

Profile scale. The numbers reported here use our quick experiment profile (Section A), with smaller replicate counts and coarser factor grids than the full design in our project plan; point estimates of bias and RMSE therefore carry more Monte Carlo noise than the full run would, though the qualitative orderings (Figure 3’s convergence-order scaling in particular) are already extremely clean at this scale.

Synthetic-data circularity. Our core claims use data generated from the same SIR model being fit, which is the correct design for isolating solver effects but cannot by itself show the model is a good description of any particular real outbreak.

Real-data provenance. The Eyam dataset (E14) was digitized from commonly-reproduced teaching tables rather than transcribed directly from the primary source in this work; see `data/raw/provenance.md`.

Single base model family. While we test SEIR, SIRS, and SIR-with-vaccination (E12), all four are still relatively low-dimensional, non-stiff systems; whether the crossover phenomenon’s specific numerical values generalize to higher-dimensional or stiffer epidemic models is untested.

7 Conclusion

Treating the ODE solver as a first-class experimental variable rather than a black box reveals structure that a black-box treatment cannot: solver-induced bias in recovered SIR parameters obeys the solver’s own convergence order almost exactly, and this bias competes with noise-induced variance in a way that can be located precisely as a crossover noise level σ^* . Below σ^* , the numerical-analysis choices this course teaches – which solver, what step size – are the dominant factor in whether an epidemiological parameter estimate is trustworthy. Above it, they are not, and the modeler’s effort is better spent on more data or a better noise model. Every figure and number in this report is generated by a single reproducible pipeline (`make all`) from the code in this repository.

References

- [1] A. Capaldi, S. Behrend, B. Berman, J. Smith, J. Wright, and A. L. Lloyd. Parameter estimation and uncertainty quantification for an epidemic model. *Mathematical Biosciences and Engineering*, 9(3):553–576, 2012.
- [2] D. Prodanov. Analytical parameter estimation of the sir epidemic model. applications to the COVID-19 pandemic. *Entropy*, 23(1):59, 2021.

A Experiment profiles

Every experiment config (`configs/*.yaml`) defines a `quick` and a `full` profile. `quick` is sized to run interactively (seconds to a few minutes per experiment); `full` matches the factorial sizes in our original project plan (e.g. E07’s solver $\times$ step size $\times$ noise $\times$ sampling interval $\times (\beta, \gamma) \times 200$ -replicate design) and is intended for an offline/overnight run. Every number in this report states which profile produced it.

B Reproducibility

`make setup` && `make experiments` && `make figures` && `make report` regenerates this PDF and every figure in it from a clean clone, using the seeds recorded in `results/manifest.json`.

C Team contributions

- **Saif Uz Zaman** — numerics core (models, solvers, linear algebra, gold standard).
- **Sayaad Muzahid Masfi** — report, presentation, and project narrative.
- **Ibtida bin Ahmed** — estimation and uncertainty quantification.
- **Sakif Naieb Raiyan** — web dashboard engineering and TypeScript/Python parity.
- **Aurchi Chowdhury** — experiment design, data provenance, and figure generation.